Closed differentiation and signed adjoints for Schwartz densities with zeros

Project Stardust
A classical formalization companion

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Abstract

For every real Schwartz density K, differentiation on Schwartz functions is densely closable in $L^2(|K|\,dx)$, without a square-integrable logarithmic derivative. We give a direct distributional construction using bounded coefficients and a weight-squared embedding, keeping the original graph closure distinct from its closed weak extension. Explicit total adjoint probes and multiplication by the sign of K identify the exact domain of the signed adjoint. Orthogonal graph projection constructs the positive Hilbert energy resolvent, its exact minimization gap and the full nonnegative self-adjoint energy operator. The logarithmic derivative is always absolutely integrable against $|K|\,dx$ but fails to be square-integrable at a simple zero. A normalized sign-changing polynomial-Gaussian example exhibits this failure while retaining all closed operators. An integrating factor also excludes polynomial first-order Stein weights for sign-changing kernels. The mechanisms are classical weighted operator and indefinite-inner-product theory; no new general closability or spectral principle is claimed.

1 The original spaces and conclusions

Fix a real Schwartz function $K \in \mathcal{S}(\mathbb{R};\mathbb{R})$ and write $w = |K|$, $\mu(dx) = w(x)\,dx$. No positivity, nonvanishing, normalization or finite-zero hypothesis on K is imposed. The measure μ is finite. Throughout,

$$H = L^2(\mu;\mathbb{C}), \qquad \langle u, v\rangle = \operatorname{Re}\int \overline{u}v\,d\mu$$

is regarded as a *real* Hilbert space. All adjoints and self-adjointness assertions below refer to this real structure, matching the formal sources. Distribution tests are complex Schwartz functions with complex bilinear distribution evaluation. The zero-density case gives the zero Hilbert space and is included.

Let D_0 take the H class of a Schwartz function f to the class of f'. Its representative independence will be proved, rather than presumed. The principal result is the following.

Theorem 1.1. *The operator D_0 is densely defined and closable. Its graph closure D is densely defined and closed. Set $Ju = \sigma u$, where $\sigma = 1$ on $\{K \geq 0\}$ and $\sigma = -1$ on $\{K < 0\}$. The signed adjoint is the densely defined closed operator*

$$D^+ = JD^*J, \qquad \operatorname{dom} D^+ = \{v \in H : Jv \in \operatorname{dom} D^*\}.$$

For the nondegenerate signed real form $[u, v] = \operatorname{Re}\int \overline{u}vK\,dx$, it satisfies

$$[Du, v] = [u, D^+v] \quad (u \in \operatorname{dom} D, v \in \operatorname{dom} D^+),$$

and this pairing condition characterizes its whole domain and value. The Hilbert energy operator

$$A = D^*D, \qquad \operatorname{dom} A = \{u \in \operatorname{dom} D : Du \in \operatorname{dom} D^*\}$$

is densely defined, closed, nonnegative and self-adjoint. Its resolvent $R = (I + A)^{-1}$ is a bounded symmetric positive contraction, injective with dense range. For every $x \in H$ and $u \in \operatorname{dom} D$,

$$\begin{aligned}\|x - u\|^2 + \|Du\|^2 = \|x - Rx\|^2 + \|DRx\|^2 \\ + \|u - Rx\|^2 + \|Du - DRx\|^2.\end{aligned} \tag{1}$$

Thus Rx is the unique minimizer of the displayed energy.

The positive operator A uses the *Hilbert* adjoint D^*, not the signed adjoint D^+. Positivity means its Hilbert quadratic form is nonnegative; it is not an assertion of order preservation or a probabilistic transition kernel. Neither operator is identified with a time generator of a signed pseudoprocess.

2 A bounded-coefficient distributional rigging

For $u \in H$ and $\varphi \in \mathcal{S}(\mathbb{R}; \mathbb{C})$ define

$$Iu(\varphi) = \int \varphi u \, d\mu, \qquad Eu = I(wu), \qquad Fu = I(bu),$$

where

$$b(x) = 2\sigma(x)K'(x), \qquad w(x)b(x) = 2K(x)K'(x). \tag{2}$$

Both w and b are bounded measurable functions, since K, K' are Schwartz. In particular these formulas define continuous real-linear maps from H into the complex tempered distributions. For example Cauchy–Schwarz gives

$$|Eu(\varphi)| \le \|u\|_H \|w\varphi\|_H, \qquad |Fu(\varphi)| \le \|u\|_H \|b\varphi\|_H,$$

and each second factor is at most a fixed constant times $\sup|\varphi|$. Thus every evaluation is continuous in u; the bounds also give the usual distribution topology continuity. No quotient by K appears.

It is important to retain the extra factor in E:

$$Eu(\varphi) = \int \varphi u \, w^2 \, dx = \int \varphi u \, K^2 \, dx. \tag{3}$$

This is not the distribution represented by u against Lebesgue measure, which need not exist locally across zeros. Its density is integrable, because $\int |u| w^2 \le \|u\|_H (\int w^3)^{1/2} < \infty$. The map E is injective. Indeed $Eu = 0$ implies $uw^2 = 0$ Lebesgue almost everywhere by separation of locally integrable densities by smooth compactly supported tests. One elementary justification of that separation is to convolve locally with a compactly supported smooth approximate identity: all convolutions are zero on smaller intervals, and converge to the density in local L^1. Hence the density is zero there, and a countable interval covering gives the assertion on $\mathbb{R}$. Since $w > 0$ μ-almost everywhere, it follows that $u = 0$ in H.

For a Schwartz input f, integration by parts gives the exact identity

$$E(f') = (Ef)' - Ff. \tag{4}$$

To check all signs, evaluate the right side on φ. It is $-\int \varphi' f K^2 - \int 2\varphi f K K'$, which equals $\int \varphi f' K^2$. Every term is absolutely integrable, and the boundary terms vanish because the factors are Schwartz. This uses the smooth weight K^2, even though $|K|$ need not be differentiable at a transverse zero.

3 The weak graph and the original derivative closure

Define

$$\mathcal{G} = \{(u, v) \in H \times H : Ev = (Eu)' - Fu\}. \tag{5}$$

It is a closed real linear subspace: for a convergent sequence (u_n, v_n) in it, continuity of each test evaluation passes the identity to the limit. If $(0, v) \in \mathcal{G}$, injectivity of E gives $v = 0$. Thus $\mathcal{G}$ is the graph of a closed partially defined operator D_{w}.

Equation (4) shows that every pair (f, f') of Schwartz classes lies in this graph. In particular if a Schwartz f is zero as an H class, then its derivative class is zero: D_0 is well-defined on the range of Schwartz functions in H. This range is dense. For clarity, finite Radon measure regularity first approximates indicators of measurable sets in L^2 by continuous compactly supported functions: choose compact and open inner/outer approximations with small measure difference and a continuous cutoff between them. Truncation and simple-function approximation reduce a general L^2 function to these indicators. Finally mollification approximates a continuous compactly supported function uniformly by smooth compactly supported functions, hence in $L^2(\mu)$ since μ is finite. Such functions are Schwartz. The same argument applies to real and imaginary parts.

Since $D_0 \subset D_{\mathrm{w}}$ and the latter has closed single-valued graph, D_0 is closable. Explicitly if $f_n \to 0$ and $f_n' \to v$ in H, their graph pairs tend to $(0, v)$ in $\mathcal{G}$, so $v = 0$. The closure of the core graph therefore defines a densely defined closed operator D with

$$D_0 \subset D \subset D_{\mathrm{w}}. \tag{6}$$

This is an inclusion, not a proved identification of the minimal graph closure with every distributional solution of (5). The Schwartz graph is a core for D by its definition. No different maximal weighted Sobolev domain or condition at individual zeros is substituted for it.

The nonnegative core form $\kappa\|D_0 f\|^2$ is closable for every $\kappa \geq 0$. For $\kappa > 0$, a sequence tending to zero whose derivative energies are Cauchy has derivatives Cauchy in H, hence convergent by completeness, and the preceding graph test makes their limit zero. For $\kappa = 0$ the assertion is immediate. This is positive *absolute-density* energy, not an energy obtained by integrating a square against the signed law.

4 Explicit adjoint probes and their totality

The real Hilbert adjoint of D is defined by

$$v \in \operatorname{dom} D^* \iff \exists z \in H\ \forall u \in \operatorname{dom} D :\ \langle v, Du\rangle = \langle z, u\rangle, \qquad D^* v = z.$$

Density of $\operatorname{dom} D$ makes z unique. Its graph is closed by passing these inner-product identities to limits. For every complex Schwartz test φ define explicit H vectors

$$p_\varphi = w\overline{\varphi}, \qquad q_\varphi = -w\overline{\varphi'} - b\overline{\varphi}. \tag{7}$$

Boundedness of the coefficients and finiteness of μ put both in H. Their normalization is

$$\langle p_\varphi, u\rangle = \operatorname{Re} Eu(\varphi), \quad \langle q_\varphi, u\rangle = -\operatorname{Re} Eu(\varphi') - \operatorname{Re} Fu(\varphi).$$

For $(u, v) \in \mathcal{G}$, (5) therefore gives $\langle p_\varphi, v\rangle = \langle q_\varphi, u\rangle$. Applied to $v = Du$, this proves

$$p_\varphi \in \operatorname{dom} D^*, \qquad D^* p_\varphi = q_\varphi. \tag{8}$$

The formal development defines these vectors by real Riesz representation; (7) is the direct coordinate realization verified by the two displayed pairings. No division by a vanishing density is needed.

Their real linear span is dense. If u is orthogonal to every p_φ, then $\operatorname{Re} Eu(\varphi) = 0$ for every φ. Testing $i\varphi$ gives the imaginary part as well, so $Eu = 0$ and injectivity gives $u = 0$. The closed span has zero orthogonal complement and is therefore H. In particular $\operatorname{dom} D^*$ is dense. These probes are total in H; they have not been asserted to be a graph core for D^*.

For completeness the bidual is the original D. A pair (u, v) is in its closed graph precisely when it is orthogonal to every vector of its graph's orthogonal complement. The latter consists of $(-D^*z, z)$ for $z \in \operatorname{dom} D^*$: orthogonality to (f, Df) is exactly the adjoint definition. Orthogonality of (u, v) to all these vectors is $\langle v, z\rangle = \langle u, D^*z\rangle$, the definition of (u, v) in the graph of D^{**}. Thus $D^{**} = D$, without replacing the original domain by $\operatorname{dom} D_{\mathrm{w}}$.

5 The signed form and its full adjoint domain

Multiplication by σ gives a bounded real-linear involution J: $J^2 = I$, $\|Ju\| = \|u\|$, and $\langle Ju, v\rangle = \langle u, Jv\rangle$. These follow pointwise from $\sigma^2 = 1$ and the fact that σ is real. The signed form

$$[u, v] = \langle Ju, v\rangle = \operatorname{Re} \int \overline{u} v K \, dx$$

is continuous and symmetric, and $[u, Ju] = \|u\|^2$. It is nondegenerate even if $[u, u] = 0$ for nonzero u; isotropy is different from degeneracy. The Hilbert space decomposes into the closed functions supported on $\{K > 0\}$ and $\{K < 0\}$, where this form is respectively positive and negative. The zero set has zero μ measure. This is the familiar real indefinite Hilbert-space construction by a fundamental symmetry.

Suppose $v, z \in H$. The identity $[Du, v] = [u, z]$ for every $u \in \operatorname{dom} D$ is equivalent, using symmetry of J, to

$$\langle Du, Jv\rangle = \langle u, Jz\rangle \quad \text{for every } u \in \operatorname{dom} D.$$

By the Hilbert adjoint definition this holds if and only if $Jv \in \operatorname{dom} D^*$ and $D^*Jv = Jz$. Multiplying by J proves the exact domain and formula $D^+v = JD^*Jv$ in Theorem 1.1. Its graph is the inverse image of the closed graph of D^* under the continuous invertible map $(v, z) \mapsto (Jv, Jz)$, so it is closed. Its domain is $J(\operatorname{dom} D^*)$, dense because J is a Hilbert isometry onto H and the adjoint domain is dense.

In particular Jp_φ is in the signed-adjoint domain with $D^+(Jp_\varphi) = Jq_\varphi$; these are actual dense test-generated vectors. The signed adjoint is not obtained by presuming a formal logarithmic derivative acts on every Schwartz input. Behavior at zeros must satisfy its actual transported Hilbert-adjoint domain.

6 Graph projection and the exact energy minimizer

The graph of D is a closed subspace of the product Hilbert space $H \oplus H$ with norm squared $\|u\|^2 + \|v\|^2$. Project $(x, 0)$ orthogonally onto it and write the projection as (Rx, Gx). Thus $Rx \in \operatorname{dom} D$ and $Gx = DRx$. Orthogonal projection and the two coordinate maps are bounded linear, so R, G are bounded real-linear maps. Orthogonality of the residual gives, for every $u \in \operatorname{dom} D$,

$$\langle Rx, u\rangle + \langle Gx, Du\rangle = \langle x, u\rangle. \tag{9}$$

The adjoint definition now yields

$$Gx \in \operatorname{dom} D^*, \qquad D^*Gx = x - Rx.$$

Hence $Rx \in \operatorname{dom} D^*D$ and $(I + D^*D)Rx = x$. Taking $u = Rx$ in (9) gives

$$\|Rx\|^2 + \|Gx\|^2 = \langle x, Rx\rangle \geq 0. \tag{10}$$

Projection is contractive, so $\|Rx\| \leq \|x\|$. Self-adjointness of the bounded graph projection also gives $\langle Rx, y\rangle = \langle x, Ry\rangle$ by testing the two product inputs $(x, 0)$ and $(y, 0)$. Thus R is symmetric and positive in the Hilbert quadratic-form sense.

If $Rx = 0$, single-valuedness of the graph gives $Gx = 0$. Equation (9) then says x is orthogonal to the dense domain of D, so $x = 0$. Thus R is injective. If y is orthogonal to its range, symmetry implies $\langle x, Ry\rangle = 0$ for every x, hence $Ry = 0$ and $y = 0$. The range of R is consequently dense.

The orthogonal residual $(x - Rx, -Gx)$ is perpendicular to $(u - Rx, Du - Gx)$. Pythagoras applied to their difference gives the exact gap (1). The last two terms there are nonnegative, so Rx minimizes $\|x - u\|^2 + \|Du\|^2$. If equality holds for another input, $\|u - Rx\|^2 = 0$ gives uniqueness. Both its domain membership and its derivative value have been established before writing this minimization statement.

7 The full Hilbert energy operator

We finish the operator part without invoking an unspecified maximal differential expression. If $u \in \operatorname{dom} D$ and $v = Du \in \operatorname{dom} D^*$, set $z = D^*v$ and $x = u + z$. For every $a \in \operatorname{dom} D$, the adjoint identity gives $\langle x, a\rangle = \langle u, a\rangle + \langle v, Da\rangle$. Comparing with (9) and testing the difference against $a = u - Rx$ yields

$$\|u - Rx\|^2 + \|v - Gx\|^2 = 0.$$

Thus $u = Rx$ and $z = x - Rx$. Conversely every Rx has exactly this factorization by the previous section. Therefore

$$\operatorname{dom} A = \operatorname{ran} R, \qquad A(Rx) = x - Rx, \qquad A = D^*D \tag{11}$$

with the precise product domain in Theorem 1.1. Injectivity of R makes this single-valued and density of its range makes A densely defined. Its quadratic form is $\langle Au, u\rangle = \|Du\|^2 \geq 0$.

Symmetry follows either from D^*D or directly from symmetry of R: $\langle x - Rx, Ry\rangle = \langle Rx, y - Ry\rangle$. To prove full self-adjointness, let $z \in \operatorname{dom} A^*$ with $A^*z = y$. Testing against Rx for arbitrary $x \in H$ gives

$$\langle x - Rx, z\rangle = \langle Rx, y\rangle.$$

Symmetry of R makes this $\langle x, z - Rz - Ry\rangle = 0$ for all x. Hence $z = R(z + y)$, so $z \in \operatorname{ran} R = \operatorname{dom} A$ and $Az = z + y - R(z + y) = y$. Thus $A^* \subseteq A$; symmetry gives the reverse inclusion. An adjoint is closed by its defining pairings, so $A = A^*$ is also closed. Equation (11) proves that $I + A$ is onto and its inverse is exactly the constructed R. This completes every operator assertion of Theorem 1.1.

The weak distributional factorization retained by these actual solutions is

$$E(DRx) = (E(Rx))' - F(Rx), \qquad D^*(DRx) = x - Rx.$$

It does not identify A with an unqualified pointwise Sturm–Liouville expression or identify an energy-semigroup law with the signed density $K\,dx$.

8 An integrable score, simple zeros and a normalized example

Define $s_K = -K'/K$ where $K \neq 0$ and set it to zero where $K = 0$, the division convention of the source. The cancellation $Ks_K = -K'$ holds Lebesgue almost everywhere, including the zero set up to a null subset. To see the last qualification, every zero where $K' \neq 0$ is isolated among zeros, by the definition of a nonzero derivative. A discrete subset of $\mathbb{R}$ is countable: choose for each isolated point a rational interval containing no other point, and use the countable rational basis. Thus $K' = 0$ at almost every zero of K. Consequently

$$\int |s_K|\, d\mu = \int |K'|\, dx < \infty. \tag{12}$$

For every Schwartz test f, ordinary integration by parts gives the actual signed Stein identity

$$\int f'K\, dx = \int f s_K K\, dx.$$

The right side is absolutely integrable since f is bounded and (12) holds. Existence of this signed integral does not imply the score belongs to H.

Suppose $K(0) = 0$ and $K'(0) \neq 0$. Continuity of K' and the derivative definition give $c, C, \delta > 0$ such that, for $0 < x < \delta$,

$$|K'(x)| \geq c, \qquad 0 < |K(x)| \leq Cx.$$

Indeed the first inequality follows by staying within half of $|K'(0)|$; the quotient $K(x)/x \to K'(0)$ gives both nonvanishing and the upper bound for sufficiently small x. Then

$$\int |s_K|^2\, d\mu \geq \int_0^\delta \frac{|K'(x)|^2}{|K(x)|}\, dx \geq \frac{c^2}{C} \int_0^\delta \frac{dx}{x} = \infty.$$

Thus a transverse zero obstructs L^2 score integrability, while all of Theorem 1.1 still holds.

For a concrete normalized signed density set

$$K_0(x) = \frac{(x + x^2)e^{-x^2}}{M}, \qquad M = \int_{\mathbb{R}} (x + x^2)e^{-x^2}\, dx > 0.$$

The odd term integrates to zero by $x \mapsto -x$. The remaining $x^2 e^{-x^2}$ is nonnegative and strictly positive on $(1, 2)$, so its integral is positive and finite. Hence K_0 is Schwartz, has integral one and is strictly negative at $x = -1/2$. Also $K_0(0) = 0$ and $K_0'(0) = 1/M \neq 0$. It follows that $s_{K_0} \in L^1(|K_0|\, dx) \setminus L^2(|K_0|\, dx)$. Its original derivative closure, dense signed adjoint and positive Hilbert energy resolvent/operator nevertheless exist. This counterexample does not say the score of every sign-changing higher-heat kernel has a simple zero; that extra zero hypothesis has not been proved or assumed here.

9 Polynomial Stein weights and higher even heat marginals

There is a different first-order obstruction that needs no simple-zero hypothesis. Suppose a real polynomial P satisfies

$$\int f'K\, dx = \int fPK\, dx \quad \text{for every real Schwartz } f.$$

Integration by parts says $\int f(K' + PK) = 0$ for all such f. The residual $K' + PK$ is itself Schwartz; choosing f to be that residual gives its squared integral zero. Continuity then gives $K' + PK = 0$ at every point. If $Q(x) = \int_0^x P(y)\,dy$, differentiation yields

$$(e^{Q(x)}K(x))' = e^{Q(x)}(K'(x) + P(x)K(x)) = 0.$$

The expression is constant on $\mathbb{R}$ by the mean value theorem. Since the exponential is strictly positive, a nonzero such K has one fixed sign. Thus a sign-changing real Schwartz density cannot have any polynomial first-order Stein weight of this form. This does not preclude the measurable L^1 score above, higher derivative identities or a calculus on a specified test dual.

For example let $p = 2m > 2$ and $t > 0$, and let k_t be the real even Schwartz kernel with angular Fourier transform $\widehat{k_t}(\xi) = e^{-t\xi^p}$. Its mass is one and its second moment is zero, by differentiating that transform twice at zero. If $k_t \geq 0$ almost everywhere, the nonnegative function $x^2 k_t(x)$ would have zero integral, forcing $k_t = 0$ away from zero almost everywhere, contradicting its unit Lebesgue mass. Thus it has a negative value; positive mass also gives a positive value. It is sign-changing, so the polynomial first-order weight is impossible. On the other hand its absolute-density space has every closed operator constructed above. The formal library uses Fourier phases $e^{2\pi i\xi x}$; rescaling angular frequency changes neither zero second moment nor sign and operator conclusions.

This is a statement about each fixed signed marginal. It gives neither a finite-variation signed path law nor predictable representation, and its positive energy is not a signed-square isometry. The separate completed multichannel test-dual calculus has different domains and must retain them.

10 Classical antecedents and exact source boundary

Weighted derivative closability is classical. De Cicco–Serra Cassano [1], Section 2.2/condition (12), recall Hamza's one-dimensional criterion. Continuous w is locally bounded below where positive, so $1/w$ is locally integrable there. Their bounded-interval relaxation results do not identify our global minimal and weak graphs. Derkach–Trunk [2], Section 2.1, give the classical $T^+ = JT^*J$ mechanism. Their coupling and spectral theorems are not used. Graph projection and D^*D are classical Hilbert theory; no priority is claimed.

The exact theorem map records frozen declarations at

`bab688bb6140cd6316c80680095a914ea0d346db`.

Riesz-probe coordinates and regrouped human proofs are labelled reconstructions. The original full-build/axiom evidence is hash-checked; no research source or original domain is changed. This formalization companion awaits independent referee review.

References

[1] V. De Cicco and F. Serra Cassano, *Relaxation and optimal finiteness domain for degenerate quadratic functionals—one dimensional case*, primary author preprint, author-hosted 2024 version. https://cvgmt.sns.it/media/doc/paper/6038/DCSC-Relaxation_22_1_2024.pdf. Section 2.2/condition (12) comparison only.

[2] V. Derkach and C. Trunk, *Coupling of definitizable operators in Kreĭn spaces*, TU Ilmenau preprint M 17/03 (2017). https://d-nb.info/1125462493/34. Section 2.1 comparison only.